

## Project Overview

Research Atlas is an intelligent research exploration platform designed to help users discover research papers, open-source implementations, and trusted domain resources from a single topic query.

The goal is to move beyond traditional keyword-based search and provide a structured research atlas powered by semantic search, ranking, and knowledge extraction.

Current example domains include:

- AI Governance
- Deepfake Detection
- Medical Image Analysis
- Computer Vision
- Machine Learning

The current backend MVP is functional and implemented in Python.

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## Current User Flow

A user enters a research topic.

Example:

AI Governance

The system automatically:

1. Retrieves relevant research papers from OpenAlex.
  2. Retrieves relevant GitHub repositories.
  3. Retrieves trusted domain-specific resources.
  4. Performs semantic ranking of papers.
  5. Builds an Atlas object.
  6. Displays the results.
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## Current Backend Architecture

### Paper Retrieval Layer

Information extracted:

- Paper title
- Authors
- Publication year
- Citation count
- Abstract

Output:

List of Paper objects.

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## Repository Retrieval Layer

Information extracted:

- Repository name
- Description
- GitHub stars
- Repository URL

Output:

List of Repository objects.

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## Trusted Resource Layer

Source:

Curated resource database

Current database contains trusted AI Governance resources from organizations such as:

- IAPP
- Governance frameworks
- Legislation trackers
- Policy resources

Output:

List of Resource objects.

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# Semantic Search Pipeline

The platform does not rely solely on keyword matching.

Instead:

1. Paper title and abstract are combined.
2. Sentence Transformer embeddings are generated.
3. Embeddings are stored in ChromaDB.
4. User queries are converted into embeddings.
5. ChromaDB performs vector similarity search.

This enables semantic retrieval based on meaning rather than exact keyword overlap.

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## Ranking System

Each paper receives an Atlas Score.

This allows the platform to prioritize papers that are most relevant to the user's topic while still considering academic impact.

The score is displayed alongside each paper.

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## Current Data Model

The backend generates an Atlas object.

Atlas contains:

Topic

Overview

Papers

Repositories

Resources

Example structure:

```
{  
  "topic": "AI Governance",  
  
  "overview": "...",  
  
  "papers": [...],  
  
  "repositories": [...],  
  
  "resources": [...]  
}
```

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## What The Frontend Will Receive

For Papers:

```
{  
  "title": "...",  
  "year": 2024,  
  "citation_count": 500,  
  "score": 87.4  
}
```

For Repositories:

```
{  
  "name": "...",  
  "stars": 1000,  
  "url": "..."  
}
```

For Resources:

```
{  
  "title": "...",  
  "url": "..."  
}
```

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# Current MVP UI

The backend currently uses Streamlit to demonstrate functionality.

This is temporary.

The goal is to replace Streamlit with a dedicated frontend.

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## Frontend Expectations

The frontend should eventually provide:

### Landing Page

Topic input field

Generate Atlas button

Example topics section

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### Atlas Dashboard

Overview section

Separate dedicated sections for each:

Generate Top Papers section

Generate Top Repositories section

Give Trusted Resources section

Atlas Scores

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### Knowledge Graph (Upcoming)

The backend will soon generate graph data in the form:

nodes = [...]

edges = [...]

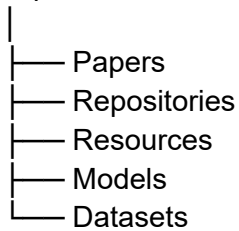
The frontend should visualize this as an interactive graph.

Potential libraries:

- React Flow
- Cytoscape.js

Expected graph structure:

Topic



Users should be able to click nodes and explore relationships.

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## Immediate Frontend Goal

Build a modern interface that can consume the Atlas object and display:

- Ranked papers
- Ranked repositories
- Trusted resources
- Atlas scores

in a clean and interactive way.

The frontend should be designed with future graph visualization and AI-powered research exploration in mind.